

One action constant from composition and spatial rotations

Result, 2026-09-27 (round 2). Independent constituent canonical algebras with constants h_i obey $[L_x, L_y] = i \sum_i h_i L_{iz}$. Requiring the **mechanical sum** $\mathbf{L} = \sum_i \mathbf{q}_i \times \mathbf{p}_i$ to obey $[L_x, L_y] = i\hbar L_z$ as an identity on the full constituent algebra forces $h_i = h$ for every constituent (Theorem 1). Equivalently, at $h \neq 0$, requiring $\mathbf{L}/h$ to generate the same spatial rotation on every constituent fixes the same normalization. Admissible pair composition propagates this equality through a connected graph. Central interactions provide a second route if conservation of the mechanical sum is imposed under a common commutator evolution (Theorem 2).

Both conclusions admit a common zero value. Spatial rotation covariance alone permits unequal h_i : for nonzero constants the dimensionless generator is $\sum_i \mathbf{L}_i/h_i$. The additional physical premise identifies this generator with the measured additive angular momentum divided by one action constant. Integer and half-integer representation labels constrain angular momenta relative to that constant; selecting its positive magnitude requires further physical input.

These are elementary conditional results, with written proofs below. They test the user's composition proposal following round 1, §§4–5 (full-read). Here h is the **reduced phase constant**, so identification with quantum mechanics would give $h = \hbar$ and $h_P = 2\pi h$. The algebra permits signed real constants; positive floors use $|h|$. All momenta are measured quantities of motion, with fixed mass and unit conventions throughout.

1. The constituent algebra and what rotation must mean

Take finitely many labelled bodies, $i = 1, \dots, N$, with masses $m_i > 0$, three spatial coordinates and three momenta. Assume an associative unital complex algebra with self-adjoint generators satisfying

$$[q_{ia}, q_{jb}] = [p_{ia}, p_{jb}] = 0, \quad [q_{ia}, p_{jb}] = i\delta_{ij}\delta_{ab}h_i 1, \quad a, b \in \{x, y, z\}. \quad (1)$$

The scalars $h_i \in \mathbb{R}$ remain fixed when the body is put in a composite. Distinct constituent algebras commute. Assume also that the normally ordered monomials $q^\alpha p^\beta$ are linearly independent: we retain the full tensor product of canonical polynomial algebras, including its commutative factors when some $h_i = 0$. This is the precise independence used in coefficient comparisons. Statistical independence of states is unnecessary; entangled states are allowed. For unbounded operator realizations all displayed identities are on a common invariant dense domain representing these polynomials faithfully.

Define, with $\epsilon_{xyz} = 1$,

$$L_{ia} = \sum_{b,c} \epsilon_{abc} q_{ib} p_{ic}, \quad L_a = \sum_i L_{ia}. \quad (2)$$

Each term in (2) has unambiguous ordering because q_{ib} and p_{ic} commute for $b \neq c$. The additivity and mechanical normalization in (2) are hypotheses. Independently rescaling each body's momentum would change its measured quantity of motion and these hypotheses.

Three distinct requirements can now be stated precisely.

- **Spatial covariance:** the map $q_i \mapsto Rq_i$, $p_i \mapsto Rp_i$ is an algebra automorphism for every $R \in SO(3)$.
- **Mechanical closure:** for one real h , $[L_a, L_b] = i\hbar \sum_c \epsilon_{abc} L_c$ on the full algebra.
- **Mechanical generator identification:** for $h \neq 0$, the generator $K_a = L_a/h$ satisfies $[K_a, q_{ib}] = i \sum_c \epsilon_{abc} q_{ic}$ and $[K_a, p_{ib}] = i \sum_c \epsilon_{abc} p_{ic}$ for every body.

The last convention gives $U(\theta) = \exp(-i\theta K_a)$ and $U(\theta)^\dagger q_{ib} U(\theta) = q_{ib} - \theta \sum_c \epsilon_{abc} q_{ic} + O(\theta^2)$, the usual active rotation of the vector's components about axis a . Integration to a group representation additionally needs a strongly continuous unitary action with these generators.

Lie commutator identities alone leave those domain and integrability obligations open. For $h \neq 0$, such an action is a representation of $SU(2)$; descent to $SO(3)$ additionally requires a full 2π rotation to act as the identity.

The round-1 phase $e^{iL\theta/h_i}$ yields (1) only after an additional canonical observable/quantization prescription. A single orbit's angular descent condition by itself supplies neither the tensor algebra nor its mechanical generator identification. The present theorem begins at that explicit algebraic extension.

2. Theorem 1: mechanical closure forces a shared constant

Theorem 1. Under (1)–(2) and the monomial independence hypothesis, mechanical closure holds for a scalar h if and only if $h_i = h$ for all i . It suffices to impose the single identity $[L_x, L_y] = ihL_z$. For $h \neq 0$, mechanical generator identification is also equivalent to $h_i = h$ for all i .

Proof. The product rule for commutators gives

$$[L_{ia}, q_{jb}] = i\delta_{ij}h_i \sum_c \epsilon_{abc}q_{ic}, \quad [L_{ia}, p_{jb}] = i\delta_{ij}h_i \sum_c \epsilon_{abc}p_{ic}. \quad (3)$$

For example, writing $L_{ix} = q_{iy}p_{iz} - q_{iz}p_{iy}$ and $L_{iy} = q_{iz}p_{ix} - q_{ix}p_{iz}$, their commutator has exactly the surviving terms

$$-ih_i q_{iy}p_{ix} + ih_i q_{ix}p_{iy} = ih_i L_{iz}.$$

Different bodies commute, so closure of the sum is equivalent to

$$\sum_i (h_i - h) L_{iz} = 0. \quad (4)$$

The monomial $q_{ix}p_{iy}$ in (4) occurs only in L_{iz} , with coefficient $h_i - h$. Independence makes that coefficient zero for each i , including a constituent with $h_i = 0$. Conversely, equal constants give closure in all three directions by the same calculation.

For generator identification, (3) gives, for example, $[L_x, q_{iy}] = ih_i q_{iz}$, whereas the required action is ihq_{iz} . Independence makes $q_{iz} \neq 0$, hence $h_i = h$. The converse follows directly from (3). $\square$

A restricted observable space needs the corresponding separation hypothesis in place of monomial independence: each relation $\sum_i c_i L_{ia} = 0$ must force every $c_i = 0$. Compression to a subspace where all orbital L_i vanish makes closure vacuous for arbitrary h_i . A single orbit, expectation value or rigidly constrained configuration can likewise lose the necessary separation. The theorem uses observable identities valid for all admitted preparations and configurations, with the constituents still individually accessible.

Transitivity. Form a graph whose vertices are bodies or species with persistent constants h_i . Put an edge between two vertices when their full pair algebra is admitted and its mechanical sum is required to have one closure constant. Theorem 1 gives equality on each edge and therefore throughout each connected component. Connectedness gives one constant for the whole class, without assuming that every pair interacts directly. Disconnected components remain unrelated unless a common composite or generator condition links them. A context-dependent h_i would require a further preparation-equivalence premise before this argument could compare contexts.

Role of three dimensions. A single planar rotation generator has $[L_z, L_z] = 0$, which tests no normalization. In three dimensions the non-abelian bracket in (4) compares the mechanical sums internally. Requiring the correct action on each q_i, p_i already compares normalizations in two dimensions by (3). Three dimensions strengthen the closure test; generator identification itself needs only one nontrivial rotation plane.

3. Rotation covariance and the interaction test

Covariance with unequal constants

For arbitrary h_i , each scalar canonical block in (1) is rotation invariant. An explicit associative realization on polynomials is

$$f * g = \mu \exp \left[\frac{i}{2} \sum_i h_i P_i \right] (f \otimes g), \quad P_i = \sum_a (\partial_{q_{ia}} \otimes \partial_{p_{ia}} - \partial_{p_{ia}} \otimes \partial_{q_{ia}}). \quad (5)$$

The exponential terminates on polynomials. The constant derivative operators commute, so either association of three factors gives the same exponential of the sum over their three pairings. Each P_i is invariant under simultaneous spatial rotations. Thus (5) supplies covariance for every tuple $(h_1, \dots, h_N)$.

If all $h_i \neq 0$, define

$$K_a = \sum_i L_{ia} / h_i.$$

Equations (3) give $[K_a, K_b] = i\epsilon_{abc}K_c$ (summed c) and the correct rotations of all constituent observables. Concretely, on $L^2(\mathbb{R}^{3N})$, $p_i = -ih_i\nabla_i$ yields $\mathbf{K} = -i\sum_i \mathbf{q}_i \times \nabla_i$, which integrates to simultaneous scalar rotations for arbitrary nonzero h_i . The identification $K_a = L_a/h$ adds exactly the normalization tested in Theorem 1. With some zero factors, spatial covariance still holds as an automorphism; commutators with their central observables vanish, so their rotations require an outer derivation.

Central interactions and conservation

For classical Newton dynamics about a fixed inertial origin, $\dot{q}_i = p_i/m_i$ and $\dot{p}_i = \sum_{j \neq i} F_{ij}$, the strong central form of the Third Law is $F_{ji} = -F_{ij}$ with F_{ij} parallel to $q_i - q_j$. It gives

$$\dot{\mathbf{L}} = \sum_{i < j} (q_i - q_j) \times F_{ij} = 0. \quad (6)$$

Equal and opposite forces together with centrality do the work here. Equation (6) uses neither a commutator nor an action constant. Transferring this conservation law to the deformed observable algebra needs a specified evolution law.

Theorem 2 (conservation propagates equality along interacting pairs). Assume (1), faithful configuration observables on open sets, and a Hamiltonian

$$H = \sum_i \frac{p_i^2}{2m_i} + \sum_{i < j} V_{ij}(r_{ij}), \quad r_{ij} = |q_i - q_j|. \quad (7)$$

Assume commutator differentiation $[p_{ia}, V] = -ih_i\partial_{q_{ia}}V$ on a common domain. This holds algebraically for polynomial potentials, or on a smooth domain away from collisions for the coordinate realization. Give time evolution one fixed nonzero action normalization h_t , $\dot{A} = [A, H]/(ih_t)$. Require conservation of the mechanical sum for every admitted pair interaction, with other pair couplings independently switchable off. For each tested edge require $V'_{ij}(r_{ij}) \neq 0$ on some open set allowing $q_i \times q_j \neq 0$. Then conservation forces $h_i = h_j$ on that edge, and therefore one value per connected interaction component.

Proof. The kinetic energies commute with each L_i . For one pair, $\nabla_j V_{ij} = -\nabla_i V_{ij}$, and centrality gives $(q_i - q_j) \times \nabla_i V_{ij} = 0$. Consequently

$$[\mathbf{L}, V_{ij}] = i(h_i - h_j)q_i \times \nabla_i V_{ij} = -i(h_i - h_j)\frac{V'_{ij}(r_{ij})}{r_{ij}}q_i \times q_j. \quad (8)$$

The configuration factor is a nonzero observable on the stated open set. Conservation, equivalent to $[\mathbf{L}, H] = 0$, forces $h_i = h_j$. When constants agree on every interacting edge, (8) proves the converse conservation statement as well. $\square$

One central harmonic interaction suffices as a polynomial test: $V_{ij} = k_{ij}|q_i - q_j|^2/2$, $k_{ij} > 0$, gives the exact defect $-ik_{ij}(h_i - h_j)q_i \times q_j$. This permits full configuration variation and keeps every step inside the polynomial algebra. Restriction to collinear configurations or to a pair's fixed centre-of-mass frame would remove this particular torque test; conservation about arbitrary fixed origins restores its stated scope. An external central potential for each isolated body gives no such exchange test and allows different constants.

Theorem 2 compares the h_i but leaves their common value relative to h_t undetermined. Demanding the full Newton equations with the physical Hamiltonian (7) gives the stronger identities

$$\dot{q}_i = \frac{h_i}{h_t} \frac{p_i}{m_i}, \quad \dot{p}_i = -\frac{h_i}{h_t} \nabla_i V. \quad (9)$$

Requiring $\dot{q}_i = p_i/m_i$ for unrestricted momenta forces $h_i = h_t$. This route already assumes a nonzero inner-commutator clock. Alternatively, assigning each body its own denominator and retaining all Newton equations must respect the shared product: for any derivation D with $Dp_{ia} = -\partial_{ia}V$,

$$0 = D[p_{ia}, p_{jb}] = i(h_i - h_j)\partial_{ia}\partial_{jb}V, \quad i \neq j. \quad (10)$$

A nonzero mixed force derivative again equates the constants. Equation (10) follows by applying the Leibniz rule to the zero cross commutator; it exposes the compatibility obligation of separate denominators. The common zero branch satisfies (10) and supports ordinary classical Hamiltonian evolution by a Poisson derivation.

4. What happens at zero, and what spin can add

Corollary (universality and positivity). Theorem 1 forces a common real value, with $h = 0$ allowed. If one constituent has an independently established $h_i \neq 0$, connected composition propagates its nonzero value to the entire component. The positivity input lies in that reference constituent. Theorem 2 likewise permits all $h_i = 0$; its commutator clock then gives trivial evolution. Classical nontrivial dynamics uses the Poisson bracket instead.

For the common family (5), all real h give associative algebras. At $h = 0$ the product is ordinary multiplication. Spatial rotations remain non-abelian: their derivations are $D_a f = \{f, L_a\}_P$, and classical angular momenta obey $\{L_a, L_b\}_P = \epsilon_{abc}L_c$. The Poisson algebra supplies the rotation structure while the multiplication algebra is commutative. For polynomial observables the common Moyal family has

$$\lim_{h \rightarrow 0} \frac{[f, g]_{*h}}{ih} = \{f, g\}_P. \quad (11)$$

Requiring a nontrivial rotation to be implemented specifically by $\exp(-i\theta L_a/h)$ selects $h \neq 0$ in the premise. Its formula requires division by h ; classical rotations remain perfectly defined by their derivations at zero. A strongly continuous unitary rotation representation alone also leaves this choice open: pullback rotates classical phase-space functions in L^2 using differential generators, while measured classical angular momenta act by multiplication.

For $h \neq 0$, supply self-adjoint angular generators integrating to a unitary $SU(2)$ representation. Its irreducible sectors have

$$J^2 = h^2 j(j+1), \quad J_z = hm, \quad j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}, \quad m = -j, -j+1, \dots, j. \quad (12)$$

Here J denotes total angular momentum, including any supplied internal spin. To recall the dimensionless argument, put $K = J/h$ and $K_{\pm} = K_x \pm iK_y$. Then $[K_z, K_{\pm}] = \pm K_{\pm}$ and $K_{\mp}K_{\pm} = K^2 - K_z^2 \mp K_z$. In an irreducible unitary sector, positivity bounds the ladder above and below. Its highest weight j has $K^2 = j(j+1)$; the lowest weight is $-j$, so $2j$ is a nonnegative integer. A full rotation acts by $e^{-2\pi im} = (-1)^{2j}$; $SO(3)$ descent retains integer j .

For ordinary scalar orbital wavefunctions, $L = q \times (-i\hbar\nabla)$ has integer orbital labels and spherical harmonics. The canonical angular algebra and this orbital spectrum are standard; see Tong, *Quantum Mechanics*, §§4.1.1–4.1.2 (passage, commutator and spherical-harmonic derivations read). Half-integer spin requires additional internal degrees of freedom or an explicitly different state space. The central-force orbital construction supplies no such degrees of freedom by itself. If independent spin observables commute with the orbital variables, requiring $J = L + S$ to rotate those variables still gives (3), hence the same orbital universality result. Requiring it also to rotate each nontrivial spin algebra fixes the spin normalization in the same way.

Equation (12) supplies dimensionless spectral ratios. The Casimir is discrete; its factor $j(j+1)$ is integer for integer j and can be fractional for half-integer j . Conditional on fixed $|h| > 0$, the smallest nonzero J^2 among all integer sectors is $2h^2$, and among all $SU(2)$ sectors it is $3h^2/4$. A chosen representation can omit these sectors; the scalar sector $j = 0$ always remains an allowed representation. Both spectral factors leave the dimensional value of h free.

More explicitly, rescaling $J \mapsto \lambda J$, $h \mapsto \lambda h$ for $\lambda > 0$ preserves the generators K , all rotation matrices, and the integer/half-integer labels. A measured nonzero magnitude ℓ and a known $j > 0$ would fix $|h| = \ell/\sqrt{j(j+1)}$, using that dimensional measurement as input. With only ℓ fixed, unbounded j allows values tending to zero. Several specified component and Casimir eigenvalues can further constrain j and their ratios; this constitutes additional spectral data. The scale freedom of the representation conditions alone remains.

5. Comparison with the two composition and dimensional results

The classical composition note, §§1–3 (full-read) concerns a fluctuation coefficient $\kappa(m) \geq 0$ and independent displacement variances. Its whole-body premise is

$$\kappa(m_1 + m_2) = \frac{m_1\kappa(m_1) + m_2\kappa(m_2)}{m_1 + m_2}.$$

For all positive masses, $f(m) = m\kappa(m)$ is additive and nonnegative, hence linear; $\kappa(m) = K \geq 0$. A reference constituent then supplies $K = m_0 u_0^2/\lambda_0 > 0$ under the stated nonzero-speed and finite-rate premises. The rotation theorem replaces this mass-additivity premise with mechanical generator identification and separation of constituent observables. A connected species graph suffices, and arbitrary positive masses need not be admitted. Both arguments propagate a coefficient; each obtains positivity only with additional input. Identifying K with the phase constant h remains another physical obligation.

There is an exact parallel in centre and relative variables. Set $M = m_1 + m_2$, $\mu = m_1 m_2 / M$,

$$R = \frac{m_1 q_1 + m_2 q_2}{M}, \quad r = q_1 - q_2, \quad P = p_1 + p_2, \quad p = \frac{m_2 p_1 - m_1 p_2}{M}.$$

Bilinearity in (1) gives

$$\begin{aligned} [R_a, P_b] &= i\delta_{ab} \frac{m_1 h_1 + m_2 h_2}{M}, & [r_a, p_b] &= i\delta_{ab} \frac{m_2 h_1 + m_1 h_2}{M}, \\ [R_a, p_b] &= i\delta_{ab} \frac{\mu}{M} (h_1 - h_2), & [r_a, P_b] &= i\delta_{ab} (h_1 - h_2). \end{aligned} \tag{13}$$

Thus canonical independence of the centre and relative algebras forces $h_1 = h_2$. The classical note has the corresponding covariance $\text{Cov}(\Delta R, \Delta r) = \Delta(\kappa_1 - \kappa_2)/M$; for its Gaussian preparations vanishing covariance gives statistical independence. Equation (13) concerns commuting

observable algebras, which can carry correlated or entangled states. The two independence premises have distinct meanings despite their matching coefficients.

The fifth-postulate note (full-read) places these results as follows. Its Theorem B obtains a single Moyal constant under full affine symplectic covariance. Applying that premise to the combined phase space already includes transformations mixing bodies, including (13), whereas separate one-body covariance allows one constant per factor as in (5). Theorem 1 isolates a narrower comparison using physical angular generators; it leaves general star-product uniqueness to the stronger covariance theorem. Theorem A retains the common zero and nonzero families under its Newton equations. Theorems C and E give conditional disturbance and concentration bounds once $|h| > 0$ is supplied. Theorem D records action rescaling. The Gaussian state restriction of B' supplies its own ζ in a commutative algebra; spatial rotations and the present commutator test leave a relation $2\zeta = |h|$ to an additional premise.

Finally, the dimensional note, Theorem A (full-read) applies to a universal floor $g(c)$ determined by fixed constants c_r . If an action product exists, its form is

$$g(c) = \left(\prod_r c_r^{a_r} \right) F(\pi_1, \dots, \pi_s), \quad \sum_r a_r d_r = (1, 2, -1). \quad (14)$$

It reduces to a pure number times the product when the fixed constants have no independent dimensionless products. Rotation angles, group structure constants, j and m are dimensionless and add no action unit to (14). When the fixed constants admit no action-dimensional product, Theorem A allows only a zero or infinite universal floor; dimensionless rotations preserve that conclusion. Treating h as a new independent fixed constant makes $|h|$ the supplied unit. Deriving a universal $|h|$ from pre-existing constants instead requires dimensional homogeneity and the same covariance argument as (14); angular universality provides no value for its dimensionless factor or proof of its positivity. For the fixed $k_e = e^2/(4\pi\epsilon_0)$, c and G of that note, this would read $|h| = Ck_e/c$. The rotation theorem leaves C undetermined. A coefficient and the infimum of a specified action observable also need a proved relation before (14) can be used as a floor statement.

6. Consequence for STATE

Item 2 gains conditional universality from mechanical angular-momentum composition: one constant per connected class follows from faithful rotation closure, or from conservation under the stated interacting commutator dynamics. The precise additional premise beyond round 1 is the identification of additive measured angular momentum with the common rotation generator. The remaining necessity task is a physical premise selecting a nonzero branch and an action unit; spin topology and Casimir discreteness leave that obligation open. All conclusions above are proved under their displayed hypotheses.